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Dramatically Enhanced Ion Conductivity of Gel Polymer Electrolyte for Supercapacitor via h-BN Nanosheets Doping

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Highlights

1. A novel h-BN-doped PVA-H_2SO_4 GPE is first fabricated using a facile method.

2. The novel GPE exhibits a dramatically enhanced ionic conductivity.
3. The h-BN nanosheets act as a super-highway for ion conducting.

Abstract: A novel hexagonal boron nitride nanosheets (h-BN)-doped poly (vinyl alcohol)-sulfuric acid (PVA-H_2SO_4) gel polymer electrolyte (GPE) is first fabricated through a facile freeze-thaw method. The ionic conductivities of the h-BN-doped PVA-H_2SO_4 GPEs with various h-BN nanosheet contents are evaluated. With the optimal content of h-BN nanosheets (0.025 mg/ml), the ionic conductivity of the GPE can reach up to 29 mS cm^{-1}, three times higher than that of GPE without the h-BN nanosheets (9 mS cm^{-1}). The h-BN nanosheets are considered as a "super-highway" for ion transport in PVA-H_2SO_4 GPE. Moreover, a symmetric qusai-solid-state supercapacitor, with activated carbon as electrodes and the h-BN nanosheets-doped PVA-H_2SO_4 gel polymer as an electrolyte and separator, delivers a high specific capacitance, good rate capability, and excellent cycle stability. At the current density of 0.5 A/g, the symmetry qusai-solid-state supercapacitor can deliver an electrode specific capacitance of 124.5 F/g, and remain 99.2 % capacitance after 5000 charge-discharge cycles. The h-BN nanosheets-doped GPE, with superior performance and facile synthesis method, appears to be a promising candidate for high-performance qusai-solid-state supercapacitors and other electrochemical devices, such as rechargeable batteries, fuel cells. This work also sheds light on the possibility to use 2D materials in advanced GPE.

Keywords: gel polymer electrolyte, boron nitride nanosheets, ionic conductivity, supercapacitor

1. Introduction

It is well known that electrolyte is a crucial part of supercapacitor. However, traditional liquid electrolytes with considerable ionic conductivity usually suffer from some inherent shortcomings, such as leakage, corrosion, and other problems. Though these problems have been tackled by replacing liquid electrolytes with solid polymer electrolytes, the application of solid polymer electrolyte in a high-performance supercapacitor is greatly limited due to its low ionic conductivity [1,2]. To achieve both advantages of the liquid electrolyte and solid electrolyte, the gel polymer electrolytes(GPEs) were proposed [3]. Mixing with polymer, salt, and solvent, GPEs with an acceptable ionic conductivity (10^{-3}-10^{-4} S cm^{-1}) are flexible, low cost, and easy to obtain. Furthermore, the use of GPE in supercapacitors can solve the leakage problem while keeping the similar electrochemical performances of the supercapacitors with a liquid electrolyte [4, 5]. Most importantly, the utilization of GPEs make the bendable power devices possible, and hence, extend the application areas of supercapacitors [6].

In the past few decades, different kinds of GPEs, such as poly (vinyl alcohol) (PVA) [7], poly (polyacrylate) (PAA) [8], poly (ethylene oxide) (PEO) [9], and poly (vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP) [10], have been synthesized and developed for supercapacitor applications. Among these GPEs, PVA attracts the interest of researchers due to its non-toxic, low cost, and good film forming property [11]. The PVA-based GPE, doped with a strong acid (H_2SO_4 or H_3PO_4) [12], strong alkaline (KOH) [13] or neutral salt (Li_2SO_4) [14] to gain the ionic conductivity, is one of the most commonly used GPEs in supercapacitors [15]. Hence, great efforts have been paid to develop high-performance quasi-solid-state supercapacitors via modifying PVA-based GPEs. One efficient way to enhance the specific capacitance of the quasi-solid-state supercapacitors is to add some redox additives, like hydroquinone [16] and indigo carmine [17], into the GPEs. When coupled with these GPEs, the quasi-solid-state supercapacitors showed a dramatical enhancement in specific capacitance. However, these redox additives, which introduce the additional pseudocapacitance, usually are harmful to the long-term cycle stability of the supercapacitors. Another way to obtain high-performance quasi-solid-state supercapacitors is to add some nanofillers into the PVA-based GPEs. For example, by adding TiO_2 or SiO_2 nanoparticles into the silicotungstic acid-PVA GPEs, the ionic conductivity of the GPEs can be maintained steadily around 10 mS cm^{-1} for more than 60 days, while the ionic conductivity fluctuates strongly with the relative humidity for the filler-free GPEs. As a result, the supercapacitors, coupled with the TiO_2 or SiO_2 nanoparticles filled GPEs, exhibit better stability [18]. Recently, 2D graphene and graphene oxides have also been introduced into GPEs. These 2D nanofillers are a benefit to decreasing the degree of crystalline in the GPEs as well as form an interconnected and continuous "super-highway" for ion movement, and hence, greatly improve the ion conductivity of these GPEs [19, 20]. As a typical, a new type of boron crosslinked graphene oxide/PVA GPE has been synthesized by Huang *et al*[21]. Due to its surface polar groups, the graphene oxide can work as an ionic conducting promoter in this GPE. As a result, there has been an obvious enhancement in specific capacitance of the GPE-based supercapacitor (about 141.8 F/g), compared with that of a supercapacitor coupled with a liquid electrolyte (109.4 F/g).

h-BN nanosheets, which are similar to graphene and can be exfoliated facilely in water through supersonication [22], have excellent chemical stability, thermal stability and intrinsic electrical insulation, compared to other 2D materials (e.g. MoS_2, graphene) [23,24]. More importantly, the intercalation of H_2SO_4 or H_3PO_4 into the h-BN through the supersonic process in water could bring lots of OH- groups on the surface of h-BN nanosheets [25, 26]. These introduced OH- groups on h-BN nanosheets could offer plenty of criss-cross "super-highways" for ion migration in GPE, and hence, make it possible for the h-BN nanosheets as a high-efficient additive to improve ionic conductivity of the GPEs.

Nevertheless, due to the poor dispersion and natural aggregation of h-BN in the polymer matrix [27], up to now, only a few works have been published on the use of h-BN nanoparticle or nanosheets as additive to improve ionic conductivity of polymer electrolyte for fuel cell applications [28-32]. For example, anhydrous Nafion/nano-BN membrane [29] and poly(vinly phosphonic acid)/sulfonated polysulfone/h-BN membrane [30] have been developed for fuel cell application. Although the ionic conductivities of these two polymer electrolytes with BN nanofillers are improved, the ionic conductivities are still lower than 0.1 mS cm^{-1}. Fortunately, some recent works have successfully demonstrated that the h-BN nanosheets can be wrapped well with the PVA chains so as to limit their aggregation and maintain a homogenous h-BN/PVA solution [33-36]. These results throw light on the possibility of obtaining high-performance PVA-based GPEs with the h-BN doping. However, to our best knowledge, there is no report of h-BN nanosheets into the PVA-H_2SO_4 GPEs for supercapacitor application.

Herein, we report the addition of h-BN nanosheets as nanofillers in PVA-H_2SO_4 GPEs and its application in a quasi-solid-state supercapacitor. With very few amount of h-BN (0.025 mg/ml) added into the PVA-H_2SO_4 GPE, the h-BN-doped PVA-H_2SO_4 GPE demonstrates the dramatical enhancement in ionic conductivity by more than 200%, compared with that of pure PVA-H_2SO_4 GPE. The dramatically enhanced ionic conductivity is attributed to the formation of ionic "super-highway" in the PVA-H_2SO_4 GPEs. Based on these exiting results, the supercapacitor, assembled with the h-BN-doped PVA-H_2SO_4 GPEs and activated carbon electrodes, exhibits excellent electrochemical performances. At the current density of 0.5 A/g, the symmetric quasi-solid-state supercapacitor coupled with h-BN-doped PVA-H_2SO_4 GPE can deliver an electrode specific capacitance of 124.5 F/g, and retain more than 99.2 % capacitance after 5000 charge-discharge cycles.

2. Experimental

2.1 Materials

All the reagents and materials, such as PVA-124 (Aladdin Co., Ltd. China), sulfuric acid (H_2SO_4, Sinopharm Chemical Reagent Co., Ltd, China), activated carbon (AC, Sinopharm

Chemical Reagent Co., Ltd, China), hexagonal boron nitride (h-BN, Aladdin Co. Ltd. China, 99.9% metal basis) were commercially available and employed without further purification.

2.2 Preparation of h-BN-doped gel polymer electrolyte

A series of BN-PVA-H_2SO_4 gel polymer electrolytes were fabricated through a simple solution-mixing/casting method. The procedure is as following. First, 0.01, 0.025, 0.05, 0.1, 0.25, and 0.5 mg/ml of h-BN suspension were obtained by respectively adding h-BN into 100 ml deionized water and ultrasonicated for 12 hours. Then 8 g PVA was dissolved into each suspension with stirring at 90°C for 2 h. After that, 10 g H_2SO_4 was added dropwisely to the solutions to form homogeneous mixtures under continuously stirring. Finally, the viscous mixtures were poured into a glass petri dish, frozen at -40°C for 12 h, and thawed at room temperature for 6 h to get the gel polymer electrolytes.

2.3 Preparation of activated carbon electrode and fabrication of quasi-solid-state supercapacitor

The AC electrode was prepared as follows: The AC (200 mg, 80%) and acetylene black (25 mg, 10%) were added into ethanol and stirred for 12 h. Then, binder polytetrafluoroethylene (25 mg, 10%) was added to the solution, and the whole slurry was groud in a mortar to obtain the AC electrode material. Then, the electrode was pressed with stainless steel net under 20 MPa. After the AC electrodes being dried at 60 °C for 24 h, the supercapacitor was fabricated by sandwiching the polymer electrolyte between two AC electrodes.

2.4 Characterization

The XRD results of h-BN powders were collected with a Co-Kα radiation (Bruker D8 Discover X-ray diffractometer). The Raman spectra of h-BN powders were collected using 532 nm laser (Renishaw inVia). The morphologies of the h-BN nanosheets were characterized by transmission electron microscope (FEI TECNAI-F30) and atomic force microscope(NT-MDT NTEGRA Spectra) in semi-contacting mode under ambient condition. The morphologies of the PVA-H_2SO_4 GPE and BN-PVA-H_2SO_4 GPE were characterized by scanning electron microscope (FEI nova nanosem 450). The PVA-H_2SO_4 GPE and BN-PVA-H_2SO_4 GPEs were tested with thermal gravimetric analysis (TGA/SDTA851, Switzerland) from 20℃ to 600℃ at a heating rate of 10℃ min^{-1} in N_2 gas. The FTIR spectra of h-BN were collected by attenuated total reflection (ATR) technique (Thermo Scientific Nicolet iS50 FT-IR spectrometer).

2.5 Electrochemical testing

All electrochemical studies were carried out at room temperature in a button cell using an electrochemical workstation system (Solarton 1470E).

Galvanostatic charge-discharge curves were measured at different current densities in the voltage range of 0 to 1.0 V. Cyclic voltammetry (CV) was carried out with a scan rate from 5 to 100 mV s^{-1} in a voltage range of 0 to 1.0 V. Electrochemical impedance spectroscopy (EIS) was carried out at open circuit potential (OCP) by applying an AC potential with 5 mV amplitude in the frequency ranges from 10^{-1} to 10^5 Hz. Moreover, cycle life was tested at a current density of 0.5 A/g for 5000 cycles.

3. Results and discussion

3.1. Preparation of BN nanosheets and BN-PVA-H_2SO_4 GPEs

Fig.1 shows the fabrication scheme of the BN-PVA-H_2SO_4 solution. The h-BN in the deionized water was exfoliated into nanosheets through the ultrasound method. The morphologies of h-BN nanosheets were characterized via TEM and AFM (XRD and Raman characterization of the h-BN were in Fig. S1). The TEM photograph exhibits the planar morphology of the h-BN nanosheets, while the AFM gives the thickness of the h-BN nanosheets was only 1-1.5 nm, corresponding to 4-5 layers. Both of them confirm the existence of nanosheets. The photographs of the BN-PVA-H_2SO_4 solutions with different h-BN contents are displayed in Fig. 1e. With the increase of the h-BN nanosheets, the PVA-H_2SO_4 solutions change from colorless and transparent to white and opacity. Meanwhile, the whole solution was homogeneous, indicating a good mixing of the h-BN nanosheets with PVA-H_2SO_4. The FESEM of the composite polymers can also confirm the conclusion (Fig. S2)

3.2. Characterization of BN-PVA-H_2SO_4 GPEs

The ionic conductivity, thermal stability, and mechanical properties of BN-PVA-H_2SO_4 GPE were investigated, and the results are shown in Fig.2. Ionic conductivity is one of the most important characteristics of the electrolyte and can be determined by EIS test through the following equations:

$$\sigma = \frac{d}{R_b \times S} \tag{1}$$

Where σ (S cm^{-1}) is the ionic conductivity of the electrolyte, d (cm) is the distance between the two stainless steel, S (cm^2) is the contace area of the electrolyte and stainless steel during test, all these data were measured several times to get the average value R_b (ohm) is the bulk resistance measured from the EIS curves. According to equation (1), the ionic conductivities of the h-BN nanosheets-doped PVA-H_2SO_4 GPEs with different contents of h-BN nanosheets are calculated and shown in Fig. 2a. Without h-BN nanosheets, the ionic conductivity of the pure PVA-H_2SO_4 GPE is only 9 mS cm^{-1}. As for the h-BN nanosheets-doped PVA-H_2SO_4 GPE, the ionic conductivity of the GPE is increased dramatically with the increase of the content of h-BN nanosheets and reaches the peak value at 0.025 mg/ml h-BN (about 29 mS cm^{-1}). It drops sharply to 1 mS cm^{-1} when the amount of h-BN nanosheets is increased to 0.1 mg/ml. This may be attributed to the restacking of the h-BN nanosheets and the blocking of the "super-highway" by the excessive h-BN nanosheets. The TGA profiles of BN-PVA-H_2SO_4 GPE and PVA-H_2SO_4 GPE from room temperature to 600℃ in N_2 gas were displayed in Fig. 2b. There seems no obviously different between the GPEs in thermal stability. It may be attributed to the tiny amount of h-BN added into the PVA-H_2SO_4 GPE. Mechanical properties determine the potential application in bendable devices. As shown in Fig. 2c-2d, the h-BN nanosheets-doped PVA-H_2SO_4 GPE display good mechanical properties. It can be easily bent and twisted without any fracture.

3.3. The electrochemical characterization of supercapacitors with 25BN-PVA-H_2SO_4 GPEs and 0BN-PVA-H_2SO_4 GPEs

To further evaluate the electrochemical performances of the supercapacitors composed of the h-BN nanosheets-doped PVA-H_2SO_4 GPEs, both the electrolytes with 0.025 mg/ml h-BN (named as 25BN-PVA-H_2SO_4) and without h-BN nanosheets (named as 0BN-PVA-H_2SO_4) are employed for comparison. In addition, all the electrodes are fabricated using commercialized active carbon (AC) as the active electrode materials.

The cyclic voltammetry curves of the supercapacitors with 0BN-PVA-H_2SO_4 GPE and 25BN-PVA-H_2SO_4 GPE were measured within the potential window from 0 to 1 V at various scanning rates and are shown in Fig. 3a and 3c. At different scanning rates, all the curves of the supercapacitors with 0BN-PVA-H_2SO_4 GPE and 25BN-PVA-H_2SO_4 GPE show a nearly rectangular shape without any redox peak, illustrating a typical electrical double layer behavior. Besides, the area of the CV curve of the supercapacitor with 25BN-PVA-H_2SO_4 GPE is larger than that of the supercapacitor with 0BN-PVA-H_2SO_4 GPE at the same scanning rate, indicating a higher capacitance of the supercapacitor with the 25BN-PVA-H_2SO_4 GPE. This result can also be verified from the following galvanostatic charge-discharge measurements.

The galvanostatic charge–discharge curves for the supercapacitors with 0BN-PVA-H_2SO_4 and 25BN-PVA-H_2SO_4 GPE measured between 0 and 1.0 V at various current densities of 0.5, 1, 2, 5 A/g are presented in Fig. 3b and 3d, respectively. The specific capacity of electrode (C_s, F/g) can be calculated as follows:

$$C_s = 4 \times \frac{I \times \Delta t}{\Delta V \times m} \tag{1}$$

where C_s is the specific capacitance of the electrode, I is charge/discharge current, Δt is the discharge time, ΔV is the potential range, and m denotes the mass of the activated carbon in two electrodes.

According to Equation (1), the electrode capacitance of the supercapacitor with 25BN-PVA-H_2SO_4 GPE is 124.5 F/g, 23.3 % higher than that of the supercapacitors without h-BN nanosheets (101 F/g) at the current density of 0.5 A/g. When the current density is at 1 A/g, 2 A/g, and 5 A/g, the electrode capacitance of the supercapacitor with 0BN-PVA-H_2SO_4 is 82 F/g, 60 F/g, and 20 F/g, respectively, only 81.2%, 59.4%, and 19.8% of the capacitance at 0.5 A/g. While the electrode capacitance of the supercapacitor with 25BN-PVA-H_2SO_4 GPE is 115 F/g, 100 F/g, and 80 F/g, respectively, 92.4%, 80.3%, and 64.0% of the capacitance at 0.5 A/g. The better rate ability of the supercapacitor with 25BN-PVA-H_2SO_4 GPE is attributed to the increased ionic conductivity by the introduction of h-BN nanosheets in GPE. And this is in agreement with the CV results.

It is known that the long-term stability is one of the most important factors to consider for the selection of an electrolyte in supercapacitor applications. The cycling stability is carried out with the galvanostatic charge-discharge measurement from 0 to 1 V at 0.5 A/g for 5000 cycles. As shown in Fig. 3f, the capacitance retention of the supercapacitor with 25BN-PVA-H_2SO_4 GPE is as high as 99.2 % after 5000 cycles with nearly 100% Coulombic efficiency, indicating that the supercapacitor has excellent electrochemical stability. Meanwhile, the supercapacitor with 25BN-PVA-H_2SO_4 GPE has a higher capacitance (123.5 F/g) than that of the supercapacitor without h-BN nanosheets (105 F/g), which indicates that the addition of h-BN nanosheets can naturally improve the capacitance of supercapacitors without any adverse effects on the cycle stability of the supercapacitor.

3.4. The role of h-BN in PVA-H_2SO_4 GPEs

To understand the role of h-BN nanosheets in the PVA-H_2SO_4 GPE, EIS technique was used to evaluate the kinetic properties of both the two GPEs in the frequency range from 100 mHz - 100 kHz. As shown in Fig. 4, the supercapacitors with both GPEs show an ideal Electric double layer capacitor (EDLC) behavior, with a small depressed semi-circle in the high-frequency area, and a slope at low-frequency end[37]. The first intersection between the curve and Z' axis denotes the bulk resistance of the supercapacitor. And the diameter of semi-circle means the charge transfer resistance of the electrode. The last slope means the diffusion resistance in the supercapacitor. The supercapacitor with the 0BN-PVA-H_2SO_4 GPE has a larger charge transfer resistance compared to that of the supercapacitor with 25BN-PVA-H_2SO_4 GPE, which means the former has a lower ionic conductivity as the electrode materials are the same in both supercapacitors. This corresponds to the improvement of ionic conductivity of the latter in Fig. 2a. The difference in resistance can be explained by the h-BN nanofillers which act as the ionic “highway” in PVA-H_2SO_4 GPE.

As a proton conducting polymer electrolyte, the conducting mechanism of PVA-H_2SO_4 GPEs is based on the Grotthuss mechanism [38]. That is, the H^+ ions transport via hopping between hydrogen bonding. Compare to the PVA-H_2SO_4 GPEs and BN-PVA-H_2SO_4 GPEs, the addition of h-BN nanosheets provide additional polar groups to facilitate the H^+ ions conduction. Besides, the rigid nanosheets compare to the soft polymer chains could be more efficient by shorting the distance of transports, just act as the role of “highway” [20], as shown in Fig. 5a and 5b. However, high content of BN nanosheets of suffers the same steric barrier effect as graphene oxides, as shown in Fig. 5c, which is mainly caused by the aggregation [19].

4. Conclusion

In summary, the h-BN nanosheets doped PVA-H_2SO_4 GPE has been fabricated and assembled with AC electrodes to construct the sandwich EDLCs. With the dramatical increase in ionic conductivity endowed by the h-BN nanosheets (from 9 to 29 mS cm^{-1} at 0.025 mg/ml), the EDLC shows excellent electrochemical performances. Contrast to the AC//PVA-H_2SO_4//AC, the AC//BN-PVA-H_2SO_4//AC device shows a higher capacitance (124.5 F/g, 23% increase) and excellent cycle life (0.8% decay after 5000 cycles). These promotions can be attributed to the ionic “super-highway” provided by the h-BN nanosheets. The existence of “super-highway” improves the ionic conductivity and thus leads to the excellent electrochemical performances of the device. These findings show that the h-BN nanosheets-doped GPE appears to be a promising candidate of high-performance quasi-solid-state supercapacitors as well as other electrochemical devices.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://

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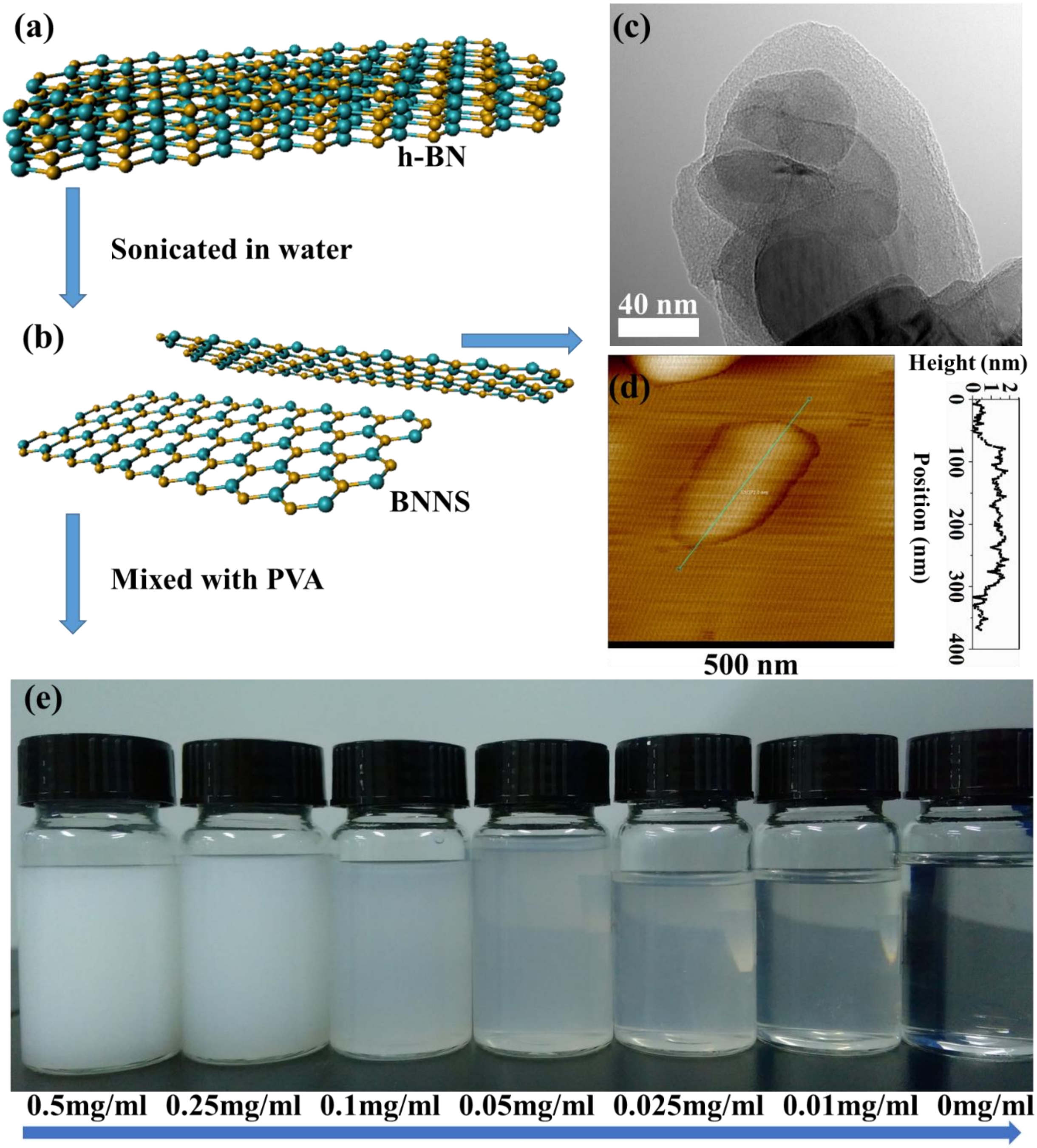

Fig.1 Schematic diagram of the BN-PVA-H_2SO_4 GPEs processing: (a-b) The scheme illustrate the fabrication of BN nanosheets; (c-d) The TEM image and AFM image of the BN nanosheets; (e) The photo of the PVA-H_2SO_4 solutions with various BN contents.

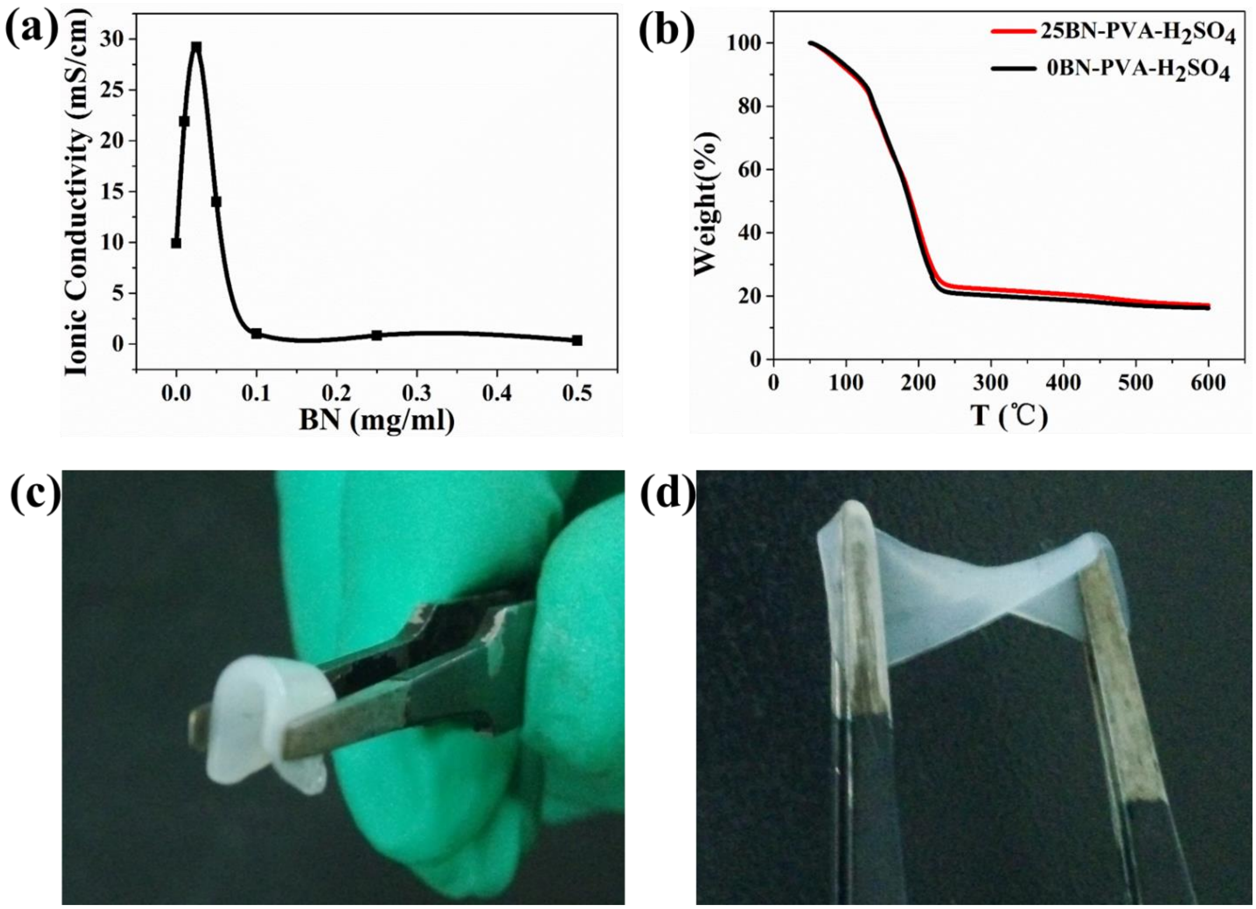

Fig.2. The properties of BN-PVA-H_2SO_4 GPEs. (a) The Ionic conductivity of BN-PVA-H_2SO_4 GPEs; (b) TGA profiles of 25BN-PVA-H_2SO_4 GPE and 0BN-PVA-H_2SO_4 GPE from room temperature to 600℃ in N_2 gas; (c-d) Photographs of 25BN-PVA-H_2SO_4 GPE in bending and twisting states.

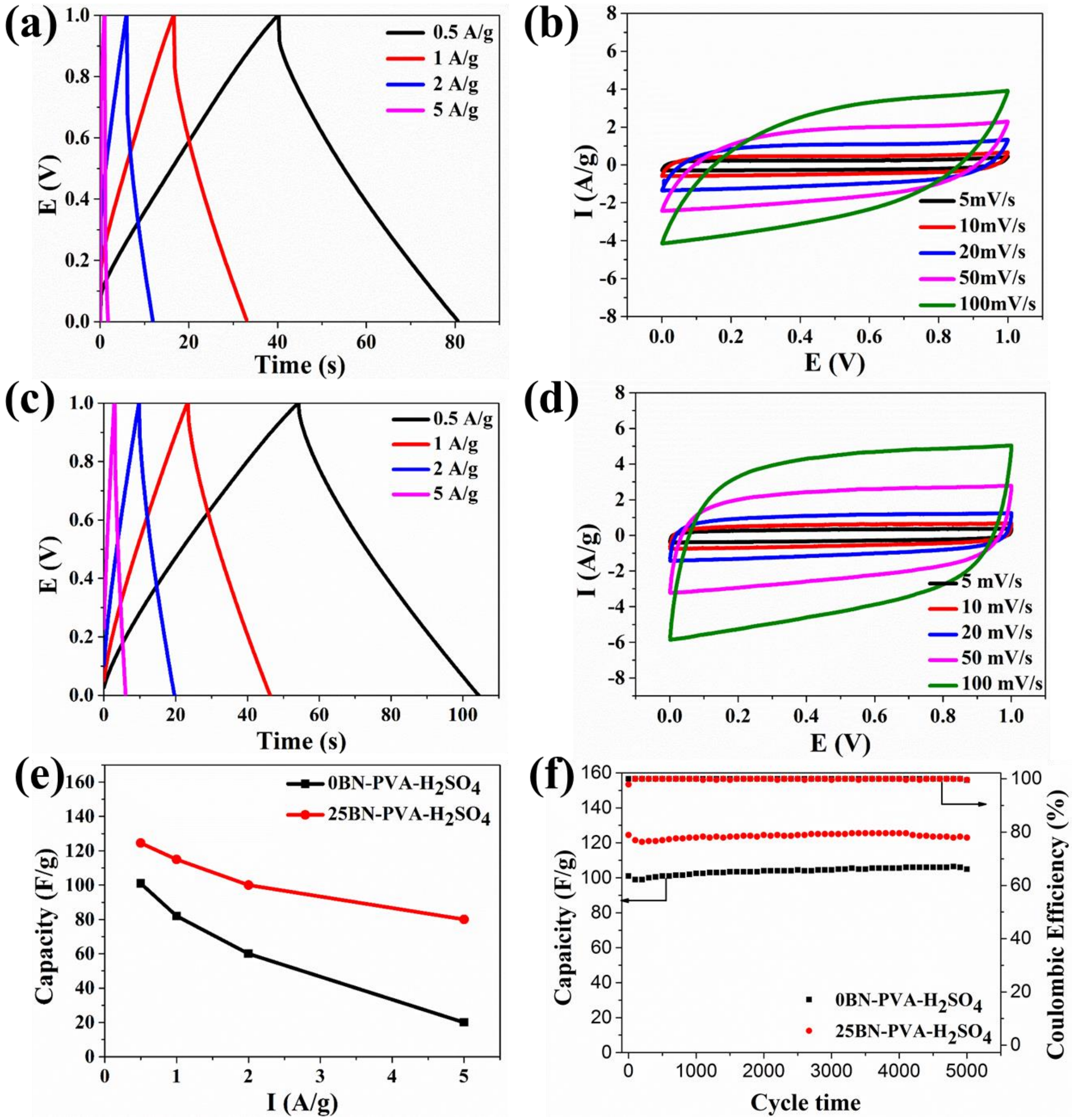

Fig.3 (a-b) The galvanostatic charge-discharge behaviors and cycle voltammograms of $0BN\text{-}PVA\text{-}H_2SO_4$ GPE; (c-d) The galvanostatic charge-discharge behaviors and cycle voltammograms of 25BN-PVA-H_2SO_4 GPE; (e-f) The rate capability, cycle stability and coulombic efficiency of 0BN-PVA-H_2SO_4 GPE and 25BN-PVA-H_2SO_4 GPE.

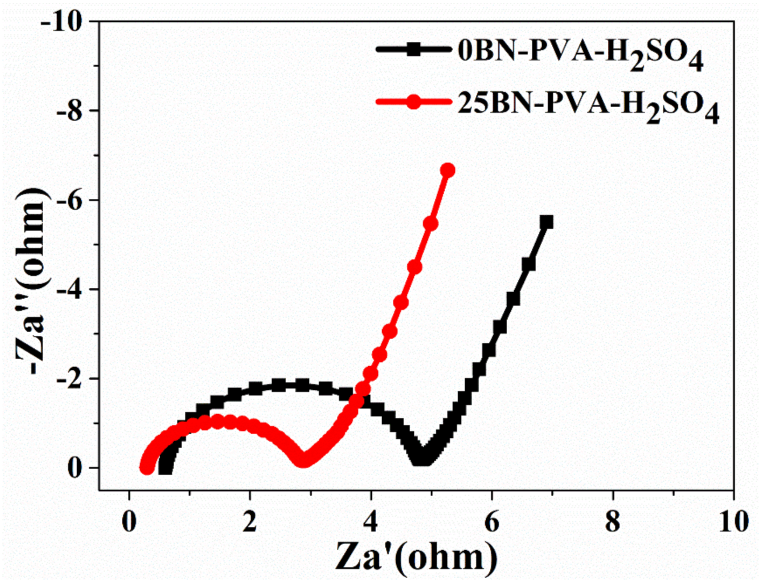

Fig.4 EIS of supercapacitors with 0BN-PVA-H_2SO_4 and 25BN-PVA-H_2SO_4 GPE.

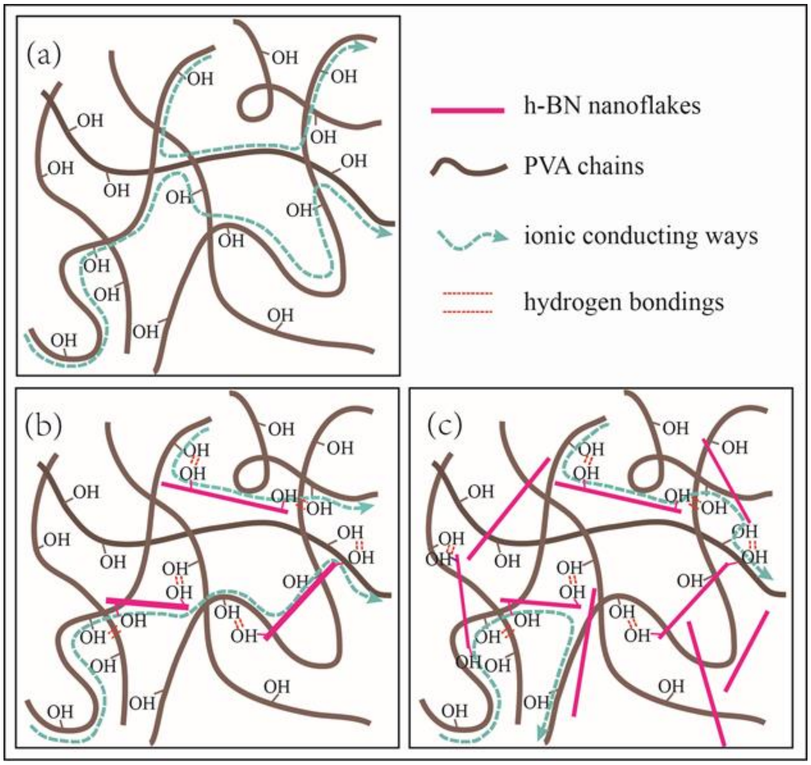

Fig.5 Illustration of ionic conduction process (a) Ions conduct along the twisted PVA chains; (b) Proper amount of BN nanosheets lead to direct conduction of ions;(c) Excessive amount of BN nanosheets hinder the conduction of ions.